

3. Repeat Exercise 1 using Algorithm 6.2. Partial pivoting

c.
$$\begin{aligned} 2x_1 - 3x_2 + 2x_3 &= 5, \\ -4x_1 + 2x_2 - 6x_3 &= 14, \\ 2x_1 + 2x_2 + 4x_3 &= 8. \end{aligned}$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 2 & 5 \\ -4 & 2 & -6 & 14 \\ 2 & 2 & 4 & 8 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} -4 & 2 & -6 & 14 \\ 2 & -3 & 2 & 5 \\ 2 & 2 & 4 & 8 \end{array} \right] \xrightarrow{\begin{array}{l} \frac{1}{2}R_1 + R_2 \\ \frac{1}{2}R_1 + R_3 \end{array}}$$

$$\max \{ |2|, |14|, |2| \} = 14$$

$$\frac{a_{21}}{a_{11}} = \frac{2}{-4} = -\frac{1}{2}, \quad \frac{a_{31}}{a_{11}} = \frac{2}{-4} = -\frac{1}{2}$$

$$\frac{1}{2}R_1 + R_2$$

$$\frac{1}{2}R_1 + R_3$$

$$\max \{ |-2|, |13| \}$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} -4 & 2 & -6 & 14 \\ 0 & 3 & 1 & 15 \\ 0 & -2 & -1 & 12 \end{array} \right]$$

$$\frac{a_{32}}{a_{22}} = \frac{-2}{3}, \quad \frac{2}{3}R_2 + R_3$$

$$\xrightarrow{\frac{2}{3}R_2 + R_3} \left[\begin{array}{ccc|c} -4 & 2 & -6 & 14 \\ 0 & 3 & 1 & 15 \\ 0 & 0 & -\frac{1}{3} & 22 \end{array} \right]$$

$$\Rightarrow -\frac{1}{3}x_3 = 22 \Rightarrow x_3 = -66$$

$$x_3 = -66$$

$$3x_2 + x_3 = 15$$

$$3x_2 - 66 = 15$$

$$\frac{3x_2}{3} = \frac{81}{3}$$

$$x_2 = 27$$

$$-4x_1 + 2x_2 - 6x_3 = 14$$

$$-4x_1 + 2(27) - 6(-66) = 14$$

$$-4x_1 + 450 = 14$$

$$\frac{-4x_1}{-4} = \frac{-436}{-4}$$

$$x_1 = 109$$

5. Repeat Exercise 1 using Algorithm 6.3. Scaled partial pivoting

c.
$$\begin{aligned} 2x_1 - 3x_2 + 2x_3 &= 5, \\ -4x_1 + 2x_2 - 6x_3 &= 14, \\ 2x_1 + 2x_2 + 4x_3 &= 8. \end{aligned}$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 2 & 5 \\ -4 & 2 & -6 & 14 \\ 2 & 2 & 4 & 8 \end{array} \right] \xrightarrow{\substack{2R_1+R_2 \\ -R_1+R_3}} \left[\begin{array}{ccc|c} 2 & -3 & 2 & 5 \\ 0 & -4 & -2 & 24 \\ 0 & 5 & 2 & 3 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 2 & -3 & 2 & 5 \\ 0 & 5 & 2 & 3 \\ 0 & -4 & -2 & 24 \end{array} \right]$$

$$s_1 = \max \{ |2|, |-3|, |2| \} = 3$$

$$\frac{|a_{22}|}{s_2} = \frac{|-4|}{6} = \frac{2}{3}$$

$$\frac{|a_{32}|}{s_3} = \frac{5}{4}$$

$$\frac{a_{32}}{a_{22}} = \frac{-4}{5}$$

$$s_2 = \max \{ |-4|, |2|, |-6| \} = 6$$

$$s_3 = \max \{ |2|, |4|, |2| \} = 4$$

$$\frac{|a_{11}|}{s_1} = \frac{2}{3}, \frac{|a_{21}|}{s_2} = \frac{4}{6} = \frac{2}{3}, \frac{|a_{31}|}{s_3} = \frac{2}{4} = \frac{1}{2}$$

$$\xrightarrow{\frac{4}{5}R_2+R_3} \left[\begin{array}{ccc|c} 2 & -3 & 2 & 5 \\ 0 & 5 & 2 & 3 \\ 0 & 0 & -\frac{2}{5} & \frac{132}{5} \end{array} \right]$$

$$\frac{a_{21}}{a_{11}} = \frac{-4}{2} = -2, \frac{a_{31}}{a_{11}} = \frac{2}{2} = 1$$

$$-\frac{5}{2} \times -\frac{2}{5}x_3 = \frac{132}{5} \times -\frac{5}{2}$$

$$x_3 = -66$$

$$5x_2 + 2x_3 = 3$$

$$5x_2 + 2(-66) = 3$$

$$5x_2 - 132 = 3$$

$$\frac{5x_2}{5} = \frac{135}{5}$$

$$x_2 = 27$$

$$2x_1 - 3x_2 + 2x_3 = 5$$

$$2x_1 - 3(27) + 2(-66) = 5$$

$$2x_1 - 213 = 5$$

$$2x_1 = 218$$

$$x_1 = 109$$

Ex 6.2 Solve by partial pivoting.

(3C) $2x_1 - 3x_2 + 2x_3 = 5$
 $-4x_1 + 2x_2 - 6x_3 = 14$
 $2x_1 + 2x_2 + 4x_3 = 8$

$$\left[\begin{array}{ccc|c} 2 & -3 & 2 & 5 \\ -4 & 2 & -6 & 14 \\ 2 & 2 & 4 & 8 \end{array} \right]$$

$\max\{|2|, |-4|, |2|\} = 4$

$R_1 \leftrightarrow R_2$
 $\rightarrow \left[\begin{array}{ccc|c} -4 & 2 & -6 & 14 \\ 2 & -3 & 2 & 5 \\ 2 & 2 & 4 & 8 \end{array} \right]$

$\frac{a_{21}}{a_{11}} = \frac{2}{-4} = -\frac{1}{2}, \frac{a_{31}}{a_{11}} = \frac{2}{-4} = -\frac{1}{2}$

$\frac{1}{2}R_1 + R_2$
 $\frac{1}{2}R_1 + R_3 \rightarrow \left[\begin{array}{ccc|c} -4 & 2 & -6 & 14 \\ 0 & -2 & -1 & 12 \\ 0 & 3 & 1 & 15 \end{array} \right]$

$\max\{|-2|, |3|\} = 3$

$R_2 \leftrightarrow R_3$
 $\rightarrow \left[\begin{array}{ccc|c} -4 & 2 & -6 & 14 \\ 0 & 3 & 1 & 15 \\ 0 & -2 & -1 & 12 \end{array} \right]$

$\frac{a_{32}}{a_{22}} = \frac{-2}{3}, R_3 = R_3 + \frac{2}{3}R_2$
 $= -2 + \frac{2}{3}(3) = -1 + \frac{2}{3}(1)$

$\frac{2}{3}R_2 + R_3$
 $\rightarrow \left[\begin{array}{ccc|c} -4 & 2 & -6 & 14 \\ 0 & 3 & 1 & 15 \\ 0 & 0 & -\frac{1}{3} & 22 \end{array} \right]$

$-\frac{1}{3}x_3 = 22 \Rightarrow x_3 = -66$

$x_3 = -66$

$3x_2 + x_3 = 15$

$3x_2 - 66 = 15$

$3x_2 = 15 + 66$

$\frac{3}{3} \frac{81}{3}$

$x_2 = 27$

Ex 6.2 Solve by partial pivoting.

(3) $-4x_1 + 2x_2 - 6x_3 = 14$
 $-4x_1 = 14 - 2x_2 + 6x_3$
 $-4x_1 = 14 - 2(27) + 6(-66)$
 $-4x_1 = 14 - 54 - 396$
 $-4x_1 = -436$
 $x_1 = 109$

$(x_1, x_2, x_3) = (109, 27, -66)$
 with 2 row interchange
 $R_1 \leftrightarrow R_2$ then $R_2 \leftrightarrow R_3$

$\frac{a_{21}}{a_{11}} = \frac{2}{-4} = -\frac{1}{2}, \frac{a_{31}}{a_{11}} = \frac{2}{-4} = -\frac{1}{2}$

$\frac{1}{2}R_1 + R_2$
 $\frac{1}{2}R_1 + R_3 \rightarrow \left[\begin{array}{ccc|c} -4 & 2 & -6 & 14 \\ 0 & -2 & -1 & 12 \\ 0 & 3 & 1 & 15 \end{array} \right]$

$\max\{|-2|, |3|\} = 3$

$R_2 \leftrightarrow R_3$
 $\rightarrow \left[\begin{array}{ccc|c} -4 & 2 & -6 & 14 \\ 0 & 3 & 1 & 15 \\ 0 & -2 & -1 & 12 \end{array} \right]$

$\frac{a_{32}}{a_{22}} = \frac{-2}{3}$

$\frac{2}{3}R_2 + R_3$
 $\rightarrow \left[\begin{array}{ccc|c} -4 & 2 & -6 & 14 \\ 0 & 3 & 1 & 15 \\ 0 & 0 & -\frac{1}{3} & 22 \end{array} \right]$

$-\frac{1}{3}x_3 = 22$

$x_3 = -66$

$3x_2 + x_3 = 15$

$3x_2 - 66 = 15$

$3x_2 = 15 + 66$

$\frac{3}{3} \frac{81}{3}$

$x_2 = 27$

5c) Solve the system of linear eqns using scaled partial pivoting.

$$\begin{aligned} 2x_1 - 3x_2 + 2x_3 &= 5 \\ -4x_1 + 2x_2 - 6x_3 &= 14 \\ 2x_1 + 2x_2 + 4x_3 &= 8 \end{aligned}$$

$$\begin{aligned} \frac{1}{2} \times 2 & \rightarrow \begin{bmatrix} 2 & -3 & 2 & | & 5 \\ -4 & 2 & -6 & | & 14 \\ 2 & 2 & 4 & | & 8 \end{bmatrix} \begin{matrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{2} \end{matrix} \\ \frac{1}{-4} \times -4 & \rightarrow \end{aligned}$$

$$s_1 = \max\{|2|, |-3|, |2|\} = 3$$

$$s_2 = \max\{|-4|, |2|, |-6|\} = 6$$

$$s_3 = \max\{|2|, |2|, |4|\} = 4$$

$$\frac{|a_{11}|}{s_1} = \frac{2}{3}, \frac{|a_{21}|}{s_2} = \frac{4}{6} = \frac{2}{3}, \frac{|a_{31}|}{s_3} = \frac{2}{4} = \frac{1}{2}$$

$$\frac{a_{11}}{a_{11}} = \frac{-4}{2} = -2, \frac{2}{2} = 1 = \frac{a_{31}}{a_{11}}$$

$$\begin{aligned} 2R_1 + R_2 & \rightarrow \begin{bmatrix} 2 & -3 & 2 & | & 5 \\ 0 & -4 & -2 & | & 24 \\ 0 & 5 & 2 & | & 3 \end{bmatrix} \\ -R_1 + R_3 & \rightarrow \end{aligned}$$

$$\frac{|a_{22}|}{s_2} = \frac{|-4|}{6} = \frac{2}{3}, \frac{|a_{32}|}{s_3} = \frac{5}{4}$$

$$R_2 \leftrightarrow R_3 \rightarrow \begin{bmatrix} 2 & -3 & 2 & | & 5 \\ 0 & 5 & 2 & | & 3 \\ 0 & -4 & -2 & | & 24 \end{bmatrix}$$

5c) Solve the system of linear eqns using scaled partial pivoting.

$$\frac{a_{32}}{a_{22}} = \frac{-4}{5}$$

$$\frac{4}{5} R_2 + R_3 \rightarrow \begin{bmatrix} 2 & -3 & 2 & | & 5 \\ 0 & 5 & 2 & | & 3 \\ 0 & 0 & -\frac{2}{5} & | & \frac{132}{5} \end{bmatrix}$$

$$-\frac{2}{5} x_3 = \frac{132}{5}$$

$$x_3 = -66$$

$$5x_2 + 2x_3 = 3$$

$$x_2 = 27$$

$$2x_1 - 3x_2 + 2x_3 = 5$$

$$x_1 = 109$$

$(x_1, x_2, x_3) = (109, 27, -66)$ with one row interchange $R_2 \leftrightarrow R_3$

$$\frac{a_{11}}{s_1}, \frac{a_{21}}{s_2}, \frac{a_{31}}{s_3}$$

whichever is bigger is on top

$$\frac{a_{22}}{s_2}, \frac{a_{32}}{s_3}$$

whichever is bigger is on top

Partial pivoting

$$\max \{ |a_{11}|, |a_{21}|, |a_{31}| \}$$

scaled partial pivoting

$$a_{11}$$

Partial pivoting

$$\max \{ |a_{11}|, |a_{21}|, |a_{31}| \} \rightarrow \text{bigger one on top}$$

Scaled partial pivoting

$$s_1, s_2, s_3$$

$$\frac{|a_{11}|}{s_1}, \frac{|a_{21}|}{s_2}, \frac{|a_{31}|}{s_3} \quad (\text{bigger one on top})$$